ECCB: Department of Ecology and Conservation Biology

ECCB 689 Modeling Ecological Niches and Species' Distributions

Lab 05: Project Occurrence Data Audit and First Model

Use the occurrence data you intend to use for your term project. Preserve the original download, document defensible filtering decisions, and connect the retained records to your audited environmental predictors. Then fit a first, deliberately simple presence–background model and explain its limitations.

Learning goals

- Build a traceable occurrence dataset and explain every exclusion.
- Map raw and retained records against your current hypothesis about M.
- Build a modeling table using occurrences and environmental background within M.
- Fit a binomial GLM with one or two continuous predictors and interpret a response curve cautiously.

What you need

Open your existing R Project. Use sf and terra, your Data Plan, project occurrence download, current M polygon, and compatible environmental layers audited in Lab 04. Keep paths relative to the project root. Complete necessary Lab 04 revisions as you prepare your project predictors. You should have files within your Project Folder names analysis, data, and output. Within them you will have the following items.

```
analysis/   scripts including 05_project_occurrences_first_model.R
data/       original and retained occurrences, predictors, M
output/     audit tables, maps, modeling table, model, response
```

The companion 05_first_model_instructor_example.R runs with the supplied course data. It demonstrates the modeling steps after vetting; it does not audit your project occurrences. Adapt its logic to your own inputs and names. Ask for help if your data are not ready.

Part A Preserve and document your occurrences

Create your own Lab 05 script

1. Create analysis/05_project_occurrences_first_model.R in your existing course R Project.
2. Type and adapt the code blocks as you work. Use your own project occurrences and justify your decisions.

3. Be sure that your script will be able to run from the top in a fresh R session so that files and objects are recreated.

Save the original download unchanged in data/. Record source/provider, dataset or download DOI when supplied, query or taxon, download date, license or access restrictions, and raw record count. Preserve source identifiers and add a row identifier if none exists. Never edit the original CSV by hand.

```
raw <- read.csv("data/your_original_download.csv")
raw$lab_row_id <- seq_len(nrow(raw))
nrow(raw)
names(raw)
summary(raw)
```

Match the reader to your format: a tab-delimited GBIF export needs read.delim(), a csv export needs read.csv(). Confirm that longitude and latitude are numeric and represent WGS 84 degrees before creating points.

Part B Audit and make explicit decisions

Inspect each issue below. Report unavailable fields as unavailable, rather than assuming they pass a check. Choose thresholds that are appropriate for your organism, environmental grain, environmental time period, and project question.

- Missing or nonnumeric coordinates and impossible longitude or latitude ranges. Zero coordinates are flags to investigate, not automatic errors.
- Repeated source IDs, identical records, repeated coordinates, and multiple records in a raster cell. Separate duplicate downloads from legitimate repeat observations.
- Coordinate uncertainty, observation year/date, taxonomy or identification, and provider quality flags.
- Geographic outliers, records outside your geographic or temporal scope, and obvious spatial clustering. Verify unusual records before deciding they are errors.

Create output/05_occurrence_decisions.csv with columns issue, records_affected, decision, and reason. Include an explicit decision for every audit category.

Create output/05_record_audit.csv with the original row/source ID, flags, retain TRUE/FALSE, and exclusion reason. Implement your decisions in R, save data/05_project_occurrences_retained.csv, and reconcile raw count = retained count + uniquely excluded count. Keep the full audit even when you remove duplicated rows.

Part C Map before and after filtering

Save output/05_occurrence_filtering_map.png. Use paired panels with the same extent, or distinct symbols, to show raw mappable records, retained records, and the current M boundary. Report how many raw records could not be mapped. Include a legend, coordinates or a geographic reference, and a useful title.

Ask: What did filtering change geographically? Where is sampling concentrated? Do the occurrence records support your current M hypothesis? A record outside M is a reason to revisit scope, identification, or M; it is not automatically an erroneous occurrence.

Checkpoint before modeling

- Your raw file remains unchanged and your retained table can be recreated from it.
- Every exclusion has a recorded reason, and all retained coordinates are valid.
- M has an explicit biological justification and a documented CRS.
- Choose one or two continuous, biologically justified project predictors. They must agree in CRS, extent, resolution, and alignment before stacking. Examine temporal compatibility and inspect missing coverage.

Part D Build the environmental contrast

Load your retained occurrence table, M polygon, and selected predictors. Use the companion example as a guide. Replace its course-data reads and its course-specific buffer with your own files and justified M. Do not copy its UTM zone or buffer distance as a universal rule.

```
library(sf)
library(terra)
set.seed(689)


# env: compatible predictor SpatRaster; m: your M polygon
m_v <- project(vect(m), crs(env))
env_m <- mask(crop(env, m_v), m_v)


# occ_sf must have its correct source CRS
occ_v <- project(vect(occ_sf), crs(env_m))
occ_values <- extract(env_m, occ_v, ID = FALSE)
```

Check every occurrence for membership in M and complete, finite predictor values. Save a modeling-preparation audit with IDs, inside_M, complete_predictors, and your decision. Diagnose NA values before deciding which records can enter this model.

Part D Sample background and assemble the table

```
bg <- spatSample(env_m, size = 1000, method = "random",
  na.rm = TRUE, as.points = TRUE, values = TRUE)

bg_values <- as.data.frame(bg)

nrow(bg_values)
```

Background represents available environments inside M. This teaching code samples raster cells; on a longitude/latitude grid, cells have different areas. Background may include cells with recorded occurrences.

Combine complete occurrence predictor rows with background predictor rows. Keep the same predictor names and units in both. Add source IDs, coordinates with documented CRS, sample_type, and presence: use 1 for occurrence and 0 for background. Save output/05_model_data.csv and report the actual number of rows in each group. The companion example shows the complete assembly code.

Part E Fit a simple GLM

Use one or two predictors with an ecological reason for inclusion. Inspect their ranges, variation, and correlation first, if you are using two. Replace predictor1 and predictor2 below with your actual column names. Use continuous predictors for this first modeling exercise.

```
first_model <- glm(presence ~ predictor1 + predictor2,
  data = model_data, family = binomial())

summary(first_model)

saveRDS(first_model, "output/05_first_model.rds")

capture.output(summary(first_model), file = "output/05_model_summary.txt")
```

Read the coefficient direction as a conditional association in this sample. Do not use p-values as an automatic predictor-selection rule. Investigate convergence warnings and very large coefficients. Reduce to one predictor if you have a problem here; do not present an unstable fit as a successful model.

The fitted response distinguishes sampled occurrence records from sampled background. Its numeric scale depends on the sampling design, including the occurrence/background ratio. Remember, it is not a literal probability of occupancy or a map of the occupied distribution.

Part F Visualize and critique the first model

Make output/05_response.png by varying one predictor across the observed modeling range and holding the other predictor at its median. Use predict(first_model, newdata = response, type = "response"). The companion example provides complete plotting code. Label the x-axis with units and the y-axis “Occurrence versus background score.”

A straight-line term in the GLM produces a monotonic logistic curve. This shape is a modeling assumption, not evidence that the organism responds monotonically. Some predictor combinations on a conditional curve may be poorly represented in the observations.

README Lab 05 prompts

1. What records did you retain, and what were your most important filtering decisions? Give raw, retained, and excluded counts and distinguish later modeling-preparation losses.
2. How did filtering alter the spatial pattern? Where is sampling uneven? Refer to your map.
3. What does your background represent, why is it restricted to M, and how was it sampled? State the actual background count and any bias concerns.
4. What does your first model estimate? Interpret one coefficient or response curve, then give at least three specific limitations and a practical next step for each. Consider sampling bias, M, filtering, temporal mismatch, grain, predictor correlation, model form, or lack of independent evaluation.

Assignment 5 submission

- Reproducible analysis/05_project_occurrences_first_model.R and unchanged original occurrence file with source documentation.
- Record audit, occurrence decision table, retained occurrence CSV, and raw-versus-retained map with M.
- Model-preparation audit, model-ready occurrence/background CSV, saved GLM, model summary, and one response figure.
- Concise README answers, acknowledged assistance or AI use, and commits pushed to your private GitHub repository. Submit the repository link through Canvas.

Restart R and run the script from the top before submission. Files must be available through the documented project structure; do not rely on objects in your workspace or personal setwd() paths.